

Sunwhi Kim, Ph.D.

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ORCID

GOOGLE SCHOLAR

GITHUB

LAB WEBSITE

WEB CV

CV SOURCE

YOUTUBE

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PROFESSIONAL APPOINTMENTS

2025–present	Assistant Professor Department of Bio-Healthcare, Hwasung Medi-Science University, Republic of Korea - Department Chair (학과장) since September 2025; principal investigator of the Neural Systems Laboratory. - Research spanning neural-circuit mechanisms of social and defensive behaviour and the cognitive human factors of human–AI interaction.
Apr 2019–Dec 2024	Postdoctoral Fellow Department of Anatomy & Neurobiology, College of Medicine, University of Tennessee Health Science Center, Memphis, TN, USA <i>Advisor: Dr. Il Hwan Kim</i> - Circuit- and network-specific dissection of the mechanisms underlying social dysfunction in Shank3 models of autism spectrum disorder. - Developed the split-Cre circuit-targeting toolkit (Addgene #187614, #187615) and the Monosynaptically-interconnected Network Module (MNM) analysis method.

EDUCATION

Mar 2013–Aug 2019	Ph.D. in Biopsychology (이학박사, Doctor of Science) Department of Psychology, Korea University <i>Advisor: Dr. June-Seek Choi · Dissertation: The functions of the amygdala sub-nuclei and prelimbic cortex in an approach-avoidance conflict situation</i>
Mar 2006–Feb 2013	B.A. in Psychology College of Liberal Arts, Department of Psychology, Korea University

PEER-REVIEWED JOURNAL ARTICLES

Published under both "Kim" and "Kimm". Each work is listed under the surname used in the original publication. * equal contribution; † corresponding author.

1. Kim YE, Kim M*, **Kim S***, Lee R*, Ujihara Y, Marquez-Wilkins EM, Jiang YH, Yang E, Kim H, Lee C, Park C†, Kim IH† (2025). Endothelial SHANK3 regulates tight junctions in the neonatal mouse blood-brain barrier through β -Catenin signaling. *Nature Communications* 16, 1407. doi:10.1038/s41467-025-56720-1
2. **Kimm S**, Kim JJ, Choi JS† (2023). The central amygdala modulates distinctive conflict-like behaviors in a naturalistic foraging task. *Frontiers in Behavioral Neuroscience* 17, 1212884. doi:10.3389/fnbeh.2023.1212884
3. **Kim S**, Kim YE, Song I, Ujihara Y, Kim N, Jiang YH, Yin HH, Lee TH, Kim IH† (2022). Neural circuit pathology driven by Shank3 mutation disrupts social behaviors. *Cell Reports* 39(10), 110906. doi:10.1016/j.celrep.2022.110906 **COVER ARTICLE**
4. **Kim S***, Kim YE*, Kim IH† (2022). Simultaneous analysis of social behaviors and neural responses in mice using round social arena system. *STAR Protocols* 3(4), 101722. doi:10.1016/j.xpro.2022.101722
5. Kim YE*, **Kim S***, Kim IH† (2022). Neural circuit-specific gene manipulation in mouse brain in vivo using split-intein-mediated split-Cre system. *STAR Protocols* 3(4), 101807. doi:10.1016/j.xpro.2022.101807
6. Kim IH†, Kim N, **Kim S**, Toda K, Catavero CM, Courtland JL, Yin HH, Soderling S† (2020). Dysregulation of the synaptic cytoskeleton in the PFC drives neural circuit pathology, leading to social dysfunction. *Cell Reports* 32(4), 107965. doi:10.1016/j.celrep.2020.107965
7. **Kimm S**, Choi JS† (2018). Sensory and motivational modulation of immediate and delayed defensive responses under dynamic threat. *Journal of Neuroscience Methods* 307, 84-94. doi:10.1016/j.jneumeth.2018.06.023
8. Lee JH, **Kimm S**, Han JS, Choi JS† (2018). Chasing as a model of psychogenic stress: characterization of physiological and behavioral responses. *Stress* 21(4), 323-332. doi:10.1080/10253890.2018.1455090

PREPRINTS & MANUSCRIPTS

1. **Kim S**, Kim SY (2026). Human factors in detecting AI-generated portraits: age, sex, device, and confidence. *arXiv*, arXiv:2603.24048. doi:10.48550/arXiv.2603.24048 **PREPRINT**

2. **Kim St**, Kim YE, Ujihara Y, Kim IH⁺ (2024). Monosynaptically-interconnected network module (MNM) approach for high-resolution brain sub-network analysis. *bioRxiv*, 2024.02.19.581007.
doi:10.1101/2024.02.19.581007 PREPRINT

CONFERENCE PROCEEDINGS

1. Jeong JH, **Kimm S**, Lee J, Choi JS⁺ (2019). Differential encoding of events during approach–avoidance conflict situation by the prelimbic cortex. *IBRO Reports* 6, S313-S314.

2. Choi JS⁺, Lee JH, **Kim S**, Han JS (2017). Validating an animal model of stress using a chasing robot. *Proceedings of the 15th Conference of the Japanese Society for Cognitive Psychology (日本認知心理学会発表論文集)*.

3. **Kim S**, Hong EH, Jie HS, Jeong JH, Choi JS⁺ (2015). Control and data acquisition of rodent experiments using Arduino-based system. *Korean Association for Laboratory Animal Science International Symposium (한국실험동물학회 학술발표대회 논문집)*, 124.

RESEARCH FUNDING

2025–2026.02	Silver digital literacy education and ESG campaign (industry–academia regional cooperation) Gyeonggi-do RISE Programme, HSMU RISE Centre · Role: Curriculum and courseware development lead · Year 1 of 3
2025.04–2026.02	Long-term neurobehavioral safety assessment of transcranial focused ultrasound neuromodulation HSMU New Faculty Research Grant · Role: Principal Investigator

TEACHING

TERM	COURSE	CODE	CR.	INSTITUTION
2026-1	Cognitive-Behavioral Experiment and Design 인지행동실험과설계	443525	3	Hwasung Medi-Science University
2026-1	Introduction to Neuroscience 뇌과학개론	500387	3	Hwasung Medi-Science University
2026-1	Introduction to Preclinical Studies 전임상개론	500386	3	Hwasung Medi-Science University
2025-2	Understanding Genetic Engineering 유전공학의이해	500500	3	Hwasung Medi-Science University
2025-2	Animal Behavior Experiment 동물행동실험	500501	3	Hwasung Medi-Science University

TERM	COURSE	CODE	CR.	INSTITUTION
2025-2	Basic Animal Experiment 기초동물실험	500498	3	Hwasung Medi-Science University
2025-1	Introduction to Neuroscience 뇌과학개론	500387	3	Hwasung Medi-Science University
2025-1	Introduction to Preclinical Studies 전임상개론	500386	3	Hwasung Medi-Science University
2025-1	General Biology 일반생물학	500384	3	Hwasung Medi-Science University
2025-1	Physiology 생리학	242519	2	Hwasung Medi-Science University (Dept. of Nursing)
2024 Spring	Guest lecture: "Autism spectrum disorder" — Developmental and Molecular Neurobiology	ANAT 825		University of Tennessee Health Science Center
2022 Spring	Guest lecture: "Autism spectrum disorder and neural circuit-based approach" — Developmental and Molecular Neurobiology	ANAT 825		University of Tennessee Health Science Center
2016-1	Workshop lecture: animal tracking and behavioral measurement with ANY-maze 행동신경과학 워크샵 — ANY-maze 사용 및 활용 방법			Korea University

INVITED TALKS & GUEST LECTURES

2026.08.06	"AI 에이전트란 무엇이며 어떤 것들이 가능한가 (What AI agents are and what they make possible)" Wecruit, invited corporate lecture · Bundang, Seongnam, Korea
2026.07.31	"AI 에이전트란 무엇이며 어떤 것들이 가능한가? — 작은 사업장을 위한 AI 활용과 청지기의 지혜 (What AI agents are and what they make possible: using AI in a small business, and the wisdom of stewardship)" Sae-Seoul CBMC, 1393rd meeting · Chunghyun Church, Seoul, Korea
2026.07.14	"AI 에이전트를 활용한 업무 자동화 (Automating everyday work with AI agents)" Hwasung Medi-Science University, 2026 all-faculty-and-staff summer workshop · ChungHo HRD Institute, Hwaseong, Korea

2025.12.02	"3D 프린터로 취미에서 창업까지: 아이디어를 현실로 만드는 기술 (From hobby to start-up with 3D printing)" HSMU 'CareerFit' programme (invited campus lecture) · Hwasung Medi-Science University, Korea
2025	"뇌 신경 회로 타겟팅 기술을 이용한 자폐 스펙트럼 장애 메커니즘 연구 (Circuit-targeting approaches to autism spectrum disorder)" 1st HSMU Colloquium, Center for Teaching and Learning · Hwasung Medi-Science University, Korea
2022	"Neural circuit-selective Shank3 mutation disrupts social behaviors" Memphis Korean Scientists Association · Memphis, TN, USA
2022	"Neural circuit-selective Shank3 mutation disrupts social behaviors" 35th US-Korea Conference (UKC) · Washington, D.C., USA
2022	"Neural circuit pathology driven by Shank3 mutation disrupts social behaviors" Neuroscience Institute Spring Seminar, University of Tennessee Health Science Center · Memphis, TN, USA
2021	"Neural circuit pathology driven by Shank3 mutation disrupts social behaviors" BK21 Seminar, Korea University · Seoul, Republic of Korea

CONFERENCE PRESENTATIONS

2024	"The miniature neural network module (MNM) approach for mapping and analyzing modular subnetworks in the brain" Kim S, Kim YE, Ujihara Y, Kim IH · Cold Spring Harbor Laboratory Neuronal Circuits Meeting · Cold Spring Harbor, NY, USA
2018	"Rat's head-withdrawal reflex depends on visual and vibrissal information and on auditory conditioning" Kim S, Lee JY, Choi JS · Korean Society of Cognitive and Biological Psychology · Korea
2016	"A modified version of Lima-Dill model predicts approach-avoidance behavior of rats facing a predator-like robot" Kim S, Jie HS, Choi JS · Society for Neuroscience (Neuroscience 2016) · San Diego, CA, USA
2016	"Amygdala regulates approach-avoidance conflict under predatory threat in rats: a model-based approach" Kim S, Choi JS · NeuroSplash (Data Blitz) · Korea

- 2015 **"Control and data acquisition of animal behavior experiments using Arduino-based system"**
Kim S, Choi JS · Korean Psychological Association Annual Conference · Seoul, Korea
- 2015 **"Control and data acquisition of rodent experiments using Arduino-based system"**
Kim S, Hong EW, Jie HS, Jeong JH, Lee JH, Choi JS · Korean Association for Laboratory Animal Science International Symposium · Incheon, Korea
- 2015 **"Central and basolateral amygdala lesions disrupt risk avoidance but spare reflexive defensive system"**
Kimm S, Choi JS · Korean Society for Brain and Neural Sciences · Korea
- 2015 **"A cost-effective system for experimental control and data acquisition using Arduino"**
Kimm S, Hong EH, Jie HS, Choi JS · Korean Society of Cognitive and Biological Psychology · Jeju, Korea
- 2014 **"Amygdala modulates approach-avoidance but not reflexive withdrawal in a semi-naturalistic conflict situation"**
Kimm SW, Kim DY, Choi JS · Annual Pavlovian Society Meeting · Seattle, WA, USA
- 2014 **"Amygdala modulates approach-avoidance but not reflexive withdrawal in semi-naturalistic conflict situation"**
Kimm S, Kim D, Choi JS · Korean Society for Brain and Neural Sciences · Seoul, Korea
- 2014 **"Partial amygdala lesions increase approach to the robot predator and impede the expression of overall defensive behaviors except for reflexive withdrawal"**
Kimm S, Kim D, Choi JS · NeuroSplash · Korea
- 2014 **"Search of sign stimulus for rats' defensive behavior using a cat-like robot"**
Kimm S, Lee YK, Choi JS · Korean Society of Cognitive and Biological Psychology · Buyeo, Korea
- 2013 **"Past experience of mild threat reduces anxiety in rats during dynamic interaction with a robot stressor"**
Kimm S, Choi JS · 16th Annual International Meeting of the Korean Society for Brain and Neural Sciences · Korea University, Seoul, Korea
- 2013 **"Lobsterbot, a possible new tool for the study of animal's risk evaluation and decision making"**
Kimm S, Lee JY, Choi JS · Korean Society for Brain and Cognitive Science Conference · Seoul National University, Seoul, Korea

2013

"Fear of predation: artificial predation-like motions by robotic agents and their effects on defensive behavior"

Kimm S, Choi JS · NeuroSplash · Korea

SOFTWARE & REGISTERED PROGRAMS

YEAR	PROGRAM	REGISTRATION	DESCRIPTION
2016	Small Animal Defensive Response Measurement and Analysis Program	C-2016-025109	Classical-conditioning experiment control and analysis software
2015	ActiMon — 24-hour small-animal activity monitoring system	C-2015-008076	Continuous 24-hour rodent activity monitoring
2015	Serial Reaction Time Task	C-2015-008009	Attention and working-memory test battery for rodents

Research, teaching and operations tools

YEAR	PROJECT	DESCRIPTION
2026	HSMU institutional research (IR) dashboard	Agent-built dashboard that assembles institutional research reporting from university data sources
2026	HSMU course CQI automation	Automates the course-level continuous quality improvement cycle — evaluation intake, report drafting and self-assessment paperwork
2026	HSMU workshop participation platform	Real-time audience participation, voting and AI-assisted synthesis app used at the 2026 all-staff workshop — phone-based input with a live projected board
2025	Real vs. AI — public portrait-discrimination web experiment	Two-alternative forced-choice web experiment used to collect the 1,664-participant dataset behind the 2026 AI-portrait study
2026	hsmu_ai_detection_public	Open stimulus set, metadata and de-identified trial-level behavioural data for the AI-generated portrait detection study
2022	Addgene plasmid deposition: split-Cre toolkit	pAAV-Ef1a-CreN-InteinN (#187614) and pAAV-Ef1a-InteinC-CreC (#187615) — circuit-specific gene-manipulation reagents distributed to the community

AI AGENT ONBOARDING & CONSULTING

2026.07–	Hwasung Medi-Science University (화성의과학대학교) Institutional automation programme (ongoing) · AI-agent-driven automation of institutional research (IR) reporting and course-level CQI workflows <i>See Software & Registered Programs for the tools built under this programme</i>
2026.08	TheAstory (디아스토리) Lecture, workshop and consulting · AI agent onboarding for business operations
2026.06	Zespa (제스파) Lecture, workshop and consulting · AI agent onboarding for business operations

MENTORING & ADVISING

2025–	11 undergraduate advisees, Dept. of Bio-Healthcare Faculty academic advisor, Hwasung Medi-Science University
2021–2024	Yusuke Ujihara, B.A. Graduate student, UTHSC
2024.05	Elizabeth Rachel Dang M.D. candidate, UTHSC
2014	Teaching assistant, Introduction to Brain Science (ISC218) Korea University International Summer Campus
2013	Teaching assistant, Learning Psychology Korea University, Dept. of Psychology

INSTITUTIONAL & PROFESSIONAL SERVICE

2025.09–present	Department Chair, Department of Bio-Healthcare 화성의과학대학교 / Hwasung Medi-Science University
2026.02	Expert content-validity reviewer, institutional core-competency scale development 화성의과학대학교
2025.10–2025.12	Member, Scholarship Review Committee 화성의과학대학교
2025	Reviewer, Liberal 101 scholarship programme 화성의과학대학교
2025.06	Member, Major Curriculum Subcommittee 화성의과학대학교

2025–	Faculty academic advisor (11 advisees), academic-probation counselling, merit-scholarship nomination 화성의과학대학교 바이오헬스케어학과
2025–	Departmental website build, admissions presentations, chairing faculty meetings, curriculum and timetable planning 화성의과학대학교 바이오헬스케어학과
2015–	Member — Korean Psychological Association; Korean Association for Laboratory Animal Science

AWARDS & HONORS

2022	Journal cover selection, Cell Reports 39(10) Cell Press
2018	Research Scholarship Korea University
2015	BK21 PLUS Scholarship Korea University
2014	Institute of Foreign Language Studies Work Scholarship Korea University
2014	BK21 PLUS Scholarship Korea University
2013	Research Assistant Scholarship Korea University
2013	BK21 PLUS Scholarship Korea University
2010	Work Scholarship Korea University
2006	Freshman Special Scholarship Korea University

CERTIFICATIONS & TRAINING

2026-02-25	Research Ethics for Principal Investigators (Science & Engineering) 국가과학기술인력개발원 (KIRD)
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2025-07-19	Laboratory Animal Technician Practical Workshop (theory module) 한국실험동물학회 인증위원회 (KALAS)
2025-05-24	Laboratory Safety Education & Training for new researchers (8 h) 화성의과학대학교
2025-04-30	Online Training for Human Subjects Researchers on the Ethical Conduct of Research 국가생명윤리정책원 (KoNIBP)
2019-02-28	Biosafety Training / Laboratory Practicum Safety Training 한국생명공학연구원 국가연구안전관리본부 (KRIBB)
2018-09-04	Biosafety Training 한국생명공학연구원 국가연구안전관리본부 (KRIBB)
2013-05-09	CITI Program — Human Research, Basic Course Collaborative Institutional Training Initiative (via Korea University)
MEDIA COVERAGE	
2022	Neural circuit pathology driven by Shank3 mutation disrupts social behaviors — 한빛사 인터뷰 한빛사 (BRIC)
2022	Autism-linked mutation disrupts brain circuit to change social behavior Spectrum News
OUTREACH	
2026–	Makers — student maker & technology club Faculty advisor. Student club for AI coding, generative media, 3D printing, Arduino and automation, run as a bridge between the lab's human–AI research and hands-on student prototyping.
PEER REVIEW	
2024, 2025	STAR Protocols (Cell Press) Four review reports, including both rounds of one 2025 manuscript
2022	Genes, Brain and Behavior
2022	Scientific Reports
2014	Behavioural Brain Research (Elsevier) two manuscripts, co-review with doctoral advisor

TECHNICAL SKILLS

AREA	DETAIL
AI Agentic Engineering	Agentic coding and multi-agent workflows, Claude Code, OpenAI Codex, Cursor, OpenClaw, Hermes, Agent-driven automation of institutional and research workflows, Prompt and context engineering, tool/MCP integration
Neural Circuit Methods	Viral circuit tracing (retrograde / anterograde / monosynaptic rabies), Split-Cre and intersectional circuit-specific gene manipulation, Optogenetics, In vivo calcium imaging (fiber photometry, miniscope), In vivo electrophysiology, Immunohistochemistry and whole-brain mapping
Behaviour	Rodent social, defensive and approach-avoidance paradigms, Semi-naturalistic robot-predator foraging tasks, Fear conditioning, extinction and context renewal, Automated tracking (DeepLabCut, ANY-maze, SAM-based multi-animal tracking)
Human Experimentation	Large-scale online psychophysics (2AFC, reaction-time modelling), IRB protocol design and human-subjects research governance, Signal detection theory
Computational	Python, R, MATLAB, Mixed-effects and Bayesian modelling, RNA-seq / differential-expression and network analysis, Machine-learning behavioural classification
Engineering	Arduino-based experiment control and data acquisition, 3D printing and rapid prototyping, Web application development (Firebase, GitHub Pages) for experiment delivery
Languages	Korean (native), English (professional working proficiency)

REFERENCES

Dr. June-Seek Choi
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Dr. Il Hwan Kim
Associate Professor, Dept. of Anatomy & Neurobiology · University of Tennessee Health Science Center · ikim9@uthsc.edu